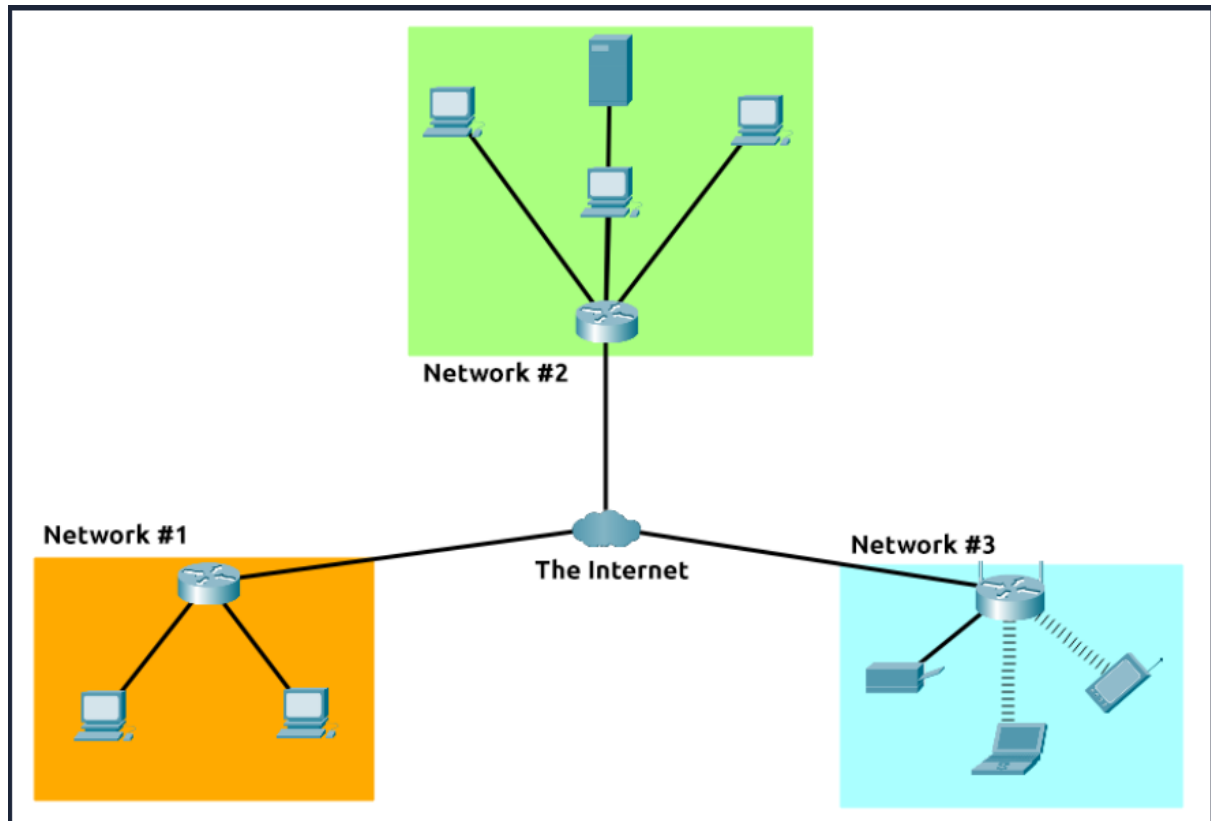


1. WHAT IS NETWORKING



Networking is the process of linking individual entities together so they can interact, exchange information, and share resources.

Depending on the domain, the term refers to two major concepts:

1. Computer Networking (Technology)

In technology, networking refers to interconnecting digital devices—such as computers, servers, smartphones, routers, and switches—using physical cables or wireless signals (Wi-Fi, cellular, Bluetooth).

- **Purpose:** Allows devices to transfer data, access shared internet connections, stream media, store files in the cloud, and use shared hardware (like printers).
- **How it works:** Devices use agreed-upon rules called **protocols** (e.g., TCP/IP, HTTP) and unique digital identifiers called **IP addresses** to send, route, and deliver data packets accurately to their destination.

Internet is made up of many small networks all joined together. These small networks are called private networks, where networks connecting these small networks are called public networks -- or the Internet! So, to recap, a network can be one of two types:

- A private network
- A public network

INTERNET

The **Internet** is a vast, global network of interconnected computers and electronic devices that communicate with each other using standardized protocols.

The **World Wide Web (WWW or "the Web")** is an information system built on top of the Internet that allows documents, web pages, and media files to be accessed and connected via hyperlinks over the web.

Invented by **Tim Berners-Lee** in **1989** at CERN, the Web transformed the raw networking infrastructure of the Internet into an accessible, user-friendly space for sharing global information.



IP ADDRESS

An **IP address** (Internet Protocol address) is a unique numerical label assigned to every device connected to a computer network that uses the Internet Protocol for communication.

Main Types of IP Addresses

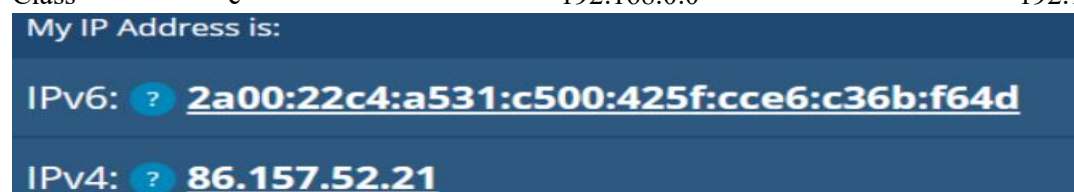
- **IPv4 vs. IPv6**
 - **IPv4:** The legacy standard using 32-bit addresses formatted as four numbers separated by dots (e.g., 192.168.1.1). It supports roughly 4.3 billion unique addresses.
 - **IPv6:** The modern standard using 128-bit hexadecimal addresses separated by colons (e.g., 2001:0db8:85a3:0000:0000:8a2e:0370:7334). It was introduced to solve IPv4 address exhaustion, offering virtually unlimited addresses.
- **Public vs. Private**
 - **Public IP:** Assigned by your Internet Service Provider (ISP) to identify your entire network on the global internet.
 - **Private IP:** Assigned internally by your router to differentiate local devices (like your phone, laptop, or smart TV) within your home or office network.

PRIVATE IP ADDRESS RANGE

Class A – 10.0.0.0—10.255.255.255

Class B – 172.16.0.0 – 172.31.255.255

Class C – 192.168.0.0 – 192.168.255.255



MAC Addresses

Devices on a network will all have a physical network interface, which is a microchip board found on the device's motherboard. This network interface is assigned a unique address at the factory it was built at, called a **MAC (Media Access Control)** address. The MAC address is a **twelve-character** hexadecimal number (*a base sixteen numbering system used in computing to represent numbers*) split into two's and separated by a colon. These colons are considered separators. For example, *a4:c3:f0:85:ac:2d*. The first six characters represent the company that made the network interface, and the last six is a unique number.



Q & A

What does the term "IP" stand for?

Internet Protocol ✓ Correct Answer

What is each section of an IP address called?

Octet ✓ Correct Answer

How many sections (in digits) does an IPv4 address have?

4 ✓ Correct Answer

What does the term "MAC" stand for?

Media Access Control ✓ Correct Answer

Deploy the interactive lab using the "View Site" button and spoof your MAC address to access the site. What is the flag?

THM{YOU_GOT_ON_TRYHACKME} ✓ Correct Answer

PING PROTOCOL

- Ping is one of the most fundamental network tools available to us. Ping uses **ICMP (Internet Control Message Protocol)** packets to determine the performance of a connection between devices, for example, if the connection exists or is reliable.
- The time taken for ICMP packets travelling between devices is measured by ping, such as in the screenshot below. This measuring is done using ICMP's echo packet and then ICMP's echo reply from the target device.
- Pings can be performed against devices on a network, such as your home network or resources like websites. This tool can be easily used and comes installed on Operating Systems (OSs) such as Linux and Windows. The syntax to do a simple ping is ping IP address or website URL

```
cmnatic@CMNatic-THM-LPTOP:~$ ping 192.168.1.254
PING 192.168.1.254 (192.168.1.254) 56(84) bytes of data:
64 bytes from 192.168.1.254: icmp_seq=1 ttl=63 time=2.18 ms
64 bytes from 192.168.1.254: icmp_seq=2 ttl=63 time=2.53 ms
64 bytes from 192.168.1.254: icmp_seq=3 ttl=63 time=2.15 ms
64 bytes from 192.168.1.254: icmp_seq=4 ttl=63 time=3.34 ms
64 bytes from 192.168.1.254: icmp_seq=5 ttl=63 time=10.3 ms
64 bytes from 192.168.1.254: icmp_seq=6 ttl=63 time=4.45 ms
^C
--- 192.168.1.254 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5008ms
rtt min/avg/max/mdev = 2.152/4.160/10.313/2.864 ms
cmnatic@CMNatic-THM-LPTOP:~$
```

Q&A

What protocol does ping use?

ICMP

✓ Correct Answer

What is the syntax to ping 10.10.10.10?

ping 10.10.10.10

✓ Correct Answer

What flag do you get when you ping 8.8.8.8?

THM[I_PINGED_THE_SERVER]

✓ Correct Answer

VARAD WALL

VARAD WALE